Predictive Bank Customer Churn Analysis

Executive Summary

Customer churn is a significant challenge for the banking industry because losing existing customers can affect revenue, profitability, and long-term customer relationships. Identifying customers who are likely to leave enables banks to take proactive measures, improve customer satisfaction, and develop targeted retention strategies. This project, Predictive Bank Customer Churn, applies machine learning techniques to identify customers at risk of leaving a bank and support data-driven decision-making.

The project uses a dataset containing 10,000 customer records with demographic, financial, and banking-related attributes, including age, credit score, tenure, account balance, number of products held, credit card ownership, active membership status, estimated salary, geography, and the target variable Exited. The target variable indicates whether a customer has churned (1) or remained with the bank (0).

The analysis began with data preprocessing and validation. The dataset was examined for missing values and duplicate records, and no missing values or duplicates were identified. Non-informative identifiers, such as CustomerId and Surname, were removed. Categorical variables were transformed using one-hot encoding. In addition, feature engineering was performed to derive four new variables: BalanceSalaryRatio, ProductDensity, EngagementProductInteraction, and AgeTenureInteraction. These features were designed to help the models capture relationships between customers' financial positions, product usage, engagement, age, and tenure.

Five machine learning algorithms were evaluated: Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, and XGBoost. The dataset was divided into training and testing subsets using a stratified split to preserve the original churn distribution. Model performance was assessed using accuracy, precision, recall, F1-score, confusion matrices, and ROC-AUC where available. Logistic Regression also underwent five-fold cross-validation to examine the consistency of its performance.

Among the evaluated models, Gradient Boosting achieved the highest test accuracy of 86.90% and a ROC-AUC score of approximately 0.8682. XGBoost delivered a comparable accuracy of 86.85% and a ROC-AUC score of approximately 0.8679. The Decision Tree achieved 85.75% accuracy, Random Forest achieved 83.85%, and Logistic Regression achieved 80.65%. Although Gradient Boosting performed best in overall accuracy and the reported ROC-AUC comparison, Random Forest achieved the strongest recall for churned customers at 64% and the highest churn-class F1-score of 0.62 among the reported results. This distinction is important because a bank focused on identifying as many potential churners as possible may prioritize recall alongside overall model performance.

Feature importance analysis using the Random Forest model identified Age as the most influential feature, with an importance score of 0.2705, followed by NumOfProducts at 0.1550 and Balance at 0.0786. The engineered EngagementProductInteraction and AgeTenureInteraction features also contributed to the model's predictions. These results help highlight customer attributes that may warrant further investigation, although feature importance alone does not establish that a variable causes churn or indicate the direction of its effect.

The project also includes a Streamlit-based interactive dashboard intended to make the model accessible to users. Its core capabilities include a customer churn risk calculator, churn-probability visualization, feature-importance analysis, and a what-if scenario simulator. Users can enter customer characteristics, adjust engagement and product-related values, and observe changes in the model's predicted churn probability. These predictions are intended to support analysis and customer-retention planning rather than replace business judgment.

In conclusion, the project demonstrates how machine learning can support proactive customer churn management in banking. Gradient Boosting is the recommended starting point when overall predictive performance is the main consideration, while Random Forest is a useful alternative when identifying a larger proportion of churned customers is the priority. Before deployment in a real banking environment, the selected model should be validated on current customer data, evaluated using business-specific costs of false positives and false negatives, and monitored for changes in prediction quality. The resulting insights and dashboard can help banking teams prioritize customer outreach, investigate potential retention opportunities, and make more informed, data-driven decisions.